Donor-Level Classification of Inflammatory Bowel Disease from Single-Cell RNA-seq: A Comparative Evaluation of Composition, Causal Factor Networks, and Variational Autoencoders

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Abstract

Donor-level disease classification from single-cell RNA sequencing (scRNA-seq) requires strict donor-aware cross-validation: naïve pipelines that split cells randomly conflate training and test donors, inflating reported performance through pseudoreplication. We present a donor-aware benchmark evaluating three feature representations across two independent IBD cohorts. The representations are: centered log-ratio (CLR) transformed cell type composition, GatedStructuralCFN dependency embeddings, and scVI variational autoencoder latent embeddings. The cohorts are the SCP259 ulcerative colitis atlas (UC vs. Healthy, $n = 30$ donors, 51 cell types) and the Kong 2023 Crohn's disease atlas (CD vs. Healthy, $n = 71$ donors, 55–68 cell types across three intestinal regions).

Compartment-stratified CLR composition achieves AUROC 0.956 ± 0.061 on SCP259; GatedStructuralCFN on the same features achieves 0.978 ± 0.050. In the Kong cohort, CFN achieves its best performance in the colon region (0.960 ± 0.055 after feature filtering), exceeding linear CLR (0.900 ± 0.100), while terminal ileum classification is dominated by linear models (CatBoost CLR 0.967 ± 0.075 vs. CFN 0.811 ± 0.164). Compartment-stratified scVI latent embeddings achieve 0.931 ± 0.101 (XGBoost), competitive with the best CLR linear result. (approximate estimate; scVI was trained on the full cohort before cross-validation, as noted in Section 4) Cross-dataset transfer (CD→UC, restricted to four shared cell types) achieves AUC 0.833 with XGBoost CLR; the reverse direction (UC→CD) performs at chance. CFN edge stability analysis reveals that compartment-wise composition eliminates spurious unit-sum-induced instability present in global composition (Jaccard similarity 0.026 vs. top-20 recurrence 1.0). CFN shows a consistent numerical advantage over linear models in the colon region of CD (AUROC 0.960 vs. 0.900), though no inter-method comparison reached statistical significance at $n \leq 34$ donors per region, and compartment-aware feature construction is critical for both classification performance and structural interpretability. All code and results are available at `https://github.com/Jonathan-321/sfn-scrna-study`.

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Key concepts — read this first

scRNA-seq (single-cell RNA sequencing) maps which genes are active in each individual cell of a tissue sample. For IBD research, this tells us not just which cell types are present but in what proportions, and whether those proportions shift under disease.

Donor-aware cross-validation means that every cell (and every derived feature) from a given patient is either entirely in training or entirely in test for a given fold. Without this, a model can "memorize" individual donors and report inflated accuracy that would not hold on a new patient.

CLR (centered log-ratio) transforms cell type proportions out of the simplex (where all values must sum to 1) into a standard real-valued space where ordinary classifiers apply. The transformation is $\mathrm{CLR}(p)_k = \log p_k - \frac{1}{K}\sum_j \log p_j$.

GatedStructuralCFN (Causal Factor Network) learns a sparse directed graph over the cell types, where each edge weight represents how much one cell typeś abundance predicts anotherś. The result is both a disease classifier and an interpretable dependency map.

scVI (single-cell Variational Inference) is a deep generative model that compresses the full gene expression profile of a cell into a low-dimensional "latent" vector. We aggregate these vectors per donor to get a fixed-size representation for classification.

AUROC (Area Under the ROC Curve) measures classifier performance on a 0–1 scale. A value of 0.5 is chance; 1.0 is perfect. It is insensitive to class imbalance, making it the right metric when healthy donors (54) outnumber CD donors (17) as in Kong 2023.

PR-AUC (Precision-Recall AUC) is a stricter metric under class imbalance: it penalizes false positives among positive predictions. We report it alongside AUROC as a secondary check.

Figures at a glance

Figure 1 gives a side-by-side view of every method tested, broken down by cohort and intestinal region. The most informative comparison is the colon panel of Kong 2023, where CFN (orange) outperforms all linear CLR baselines after feature filtering. The TI panel tells the opposite story: CatBoost on CLR achieves 0.967 while CFN reaches only 0.811, which is why region matters when evaluating structured models.

Figure 2 contrasts the CFN dependency matrices from global and compartment-level composition. The difference between the first two panels (SCP259 global vs. compartment) is the visual summary of the edge-stability result: the global matrix is diffuse and irreproducible across folds (Jaccard 0.026) while the compartment matrix has clean block structure that recurs perfectly in all five folds. Readers interested in the compositional geometry argument should examine these two panels together before reading Section 3.2.

Figure 3 documents cross-dataset transfer using only the four cell types shared between atlases. The directional asymmetry is the main takeaway: CD→UC reaches AUC 0.833 while UC→CD sits at chance. The heatmap on the right of the figure makes this asymmetry immediately visible and also shows that CFN and linear models perform similarly in this limited four-type regime.

1 Introduction

Single-cell RNA sequencing has transformed our ability to characterize the cellular architecture of complex tissues, and its application to inflammatory bowel disease (IBD) has produced high-resolution atlases of colonic and ileal mucosa under both healthy and diseased conditions [Smillie

et al., 2019, Kong et al., 2023]. A natural downstream question is whether the cellular composition of these tissues—the relative abundance of each cell type across the full atlas—is sufficient to classify individual patients by disease status, and whether structured models that capture dependencies between cell types add predictive value beyond simple linear classifiers.

Cell type composition predicts IBD status at the donor level [Smillie et al., 2019], and donor-level features derived from scRNA-seq reach competitive AUROC values against traditional clinical markers. However, published evaluations frequently suffer from a methodological gap: cells are split randomly into training and test sets without regard to donor identity, so test cells from a given donor are evaluated by a model that has seen training cells from the same donor. Because cells from the same donor are far more similar to each other than to cells from a different donor, this cross-donor leakage inflates reported performance and can produce AUROC values approaching 1.0 that would not generalize to new patients.

In parallel, structured models such as StructuralCFN [Li, 2026] have been proposed as tools for learning sparse dependency networks over features, with the dependency matrix providing interpretable structure beyond what a scalar prediction delivers. Whether such models, which carry substantially more parameters and inductive bias than logistic regression or a linear SVM, provide a performance benefit over carefully constructed linear baselines on compositional IBD data has not been established in a rigorous donor-aware evaluation.

The present study closes both gaps. We implement a strict donor-aware cross-validation protocol in which fold assignment is done at the donor level, no cell (or derived feature) from a test donor appears in any training set, and all preprocessing steps (CLR normalization, HVG selection for scVI) are fit on training donors only and applied to test donors. We then benchmark, across two independent cohorts, five classifiers (logistic regression, linear SVM, ElasticNet, XGBoost, and CatBoost) on three feature representations: (1) CLR-transformed cell type composition; (2) GatedStructuralCFN embeddings; and (3) scVI latent embeddings. We additionally evaluate cross-dataset transfer between the UC and CD cohorts using the four cell types present in both atlases, and assess the structural stability of the learned CFN dependency matrices across cross-validation folds.

Our central findings are: (i) compartment-stratified composition features with simple linear classifiers establish a strong baseline that is difficult to improve upon in the SCP259 UC cohort; (ii) CFN shows a numerical discriminative advantage in the colon region of the Kong CD cohort (AUROC 0.960 vs. 0.900 for the best linear baseline), though no inter-method comparison reached significance at $n \leq 34$ donors per region; composition remodeling in the colon appears more co-dependent than in the terminal ileum; (iii) scVI latent embeddings match linear composition performance when compartment structure is preserved in the aggregation step; (iv) CFN dependency structure is only reproducible across folds when composition is computed within compartments rather than globally; and (v) cross-dataset transfer from CD to UC is feasible with four shared cell types (AUC 0.833) but the reverse direction performs at chance, reflecting cohort size and phenotypic diversity asymmetries.

2 Methods

2.1 Datasets

2.1.1 SCP259 (Smillie et al. 2019: UC vs. Healthy)

Single-cell RNA-seq data were obtained from the Broad Single Cell Portal study SCP259 [Smillie et al., 2019]. The dataset comprises colonic biopsy specimens from 18 ulcerative colitis patients and 12 healthy controls ($n = 30$ donors). Raw count matrices were provided as gene-sorted MTX files separated by tissue compartment: epithelial (Epi), fibroblast/stromal (Fib), and immune (Imm). Cell type annotations (51 types) and donor metadata were obtained from the accompanying `all.meta2.txt` file.

Donor-level cell type composition was computed as $p_{d,k} = c_{d,k}/N_d$ where $c_{d,k}$ is the cell count for type k in donor d and N_d is the total cell count. Separate composition tables were constructed for the combined dataset (51 types) and for each compartment independently; the union of within-compartment cell types yielded a 102-dimensional compartment-stratified representation. Five-fold cross-validation splits were stratified by disease status and fixed before any model training.

2.1.2 Kong et al. 2023 (CD vs. Healthy)

Data were obtained from Kong et al. [Kong et al., 2023], accessed via NCBI GEO (GSE202021). The dataset comprises intestinal biopsy specimens from 17 Crohn's disease patients and 54 healthy controls ($n = 71$ donors) across six H5AD files. Donor-level composition tables were built from cell metadata extracted in backed mode, yielding three region-stratified tables: all regions combined (71 donors, 68 types), terminal ileum only (42 donors, 61 types), and colon only (34 donors, 55 types).

2.2 Feature Engineering

We applied two cleaning steps to the raw composition tables before model training, following standard practice for high-dimensional compositional data with sparse cell types.

Rare-type filter. Cell types present in fewer than 20% of donors (i.e., zero cells observed in that donor) were removed. These types are structurally absent in most donors; after CLR transformation they become imputed zeros and carry no discriminative signal.

Low-variance filter. Cell types with standard deviation < 0.005 in the raw proportion space across all donors were removed. The Kong 2023 composition tables contain 20–25 near-constant features per region (e.g., DC1, ILCs, Pericyte subtypes) with $\sigma < 0.005$; these contribute noise to classifier fitting and inflate effective dimensionality relative to n. After filtering, Kong feature dimensionality reduced from 68/61/55 to 43/41/32 for the all/TI/colon regions respectively. SCP259 composition (51 types, 1 sparse type) passed filtering largely intact.

2.3 CLR Transformation

Cell type composition vectors lie in the simplex Δ^{K-1} and carry a unit-sum constraint. We applied the centered log-ratio (CLR) transform [Aitchison, 1982]:

$$\mathrm{CLR}(p)_k = \log\left(\frac{p_k}{\left(\prod_{j=1}^{K} p_j\right)^{1/K}}\right) = \log(p_k) - \frac{1}{K}\sum_{j=1}^{K}\log(p_j)$$

We added a pseudocount $\varepsilon = 0.5/K$ before log transformation. CLR parameters were fit on training donors only within each fold and applied to test donors separately to prevent leakage.

2.4 Baseline Classifiers

All models were evaluated in 5-fold stratified cross-validation (fixed splits, `random_state=42`). Primary metric: AUROC. Secondary: PR-AUC.

- **LogReg**: `LogisticRegression`, L2, lbfgs, `max_iter=5000`
- **LinearSVM**: `SVC(kernel='linear', probability=True)`
- **ElasticNet**: `LogisticRegression(penalty='elasticnet', l1_ratio=0.5, solver='saga')`
- **XGBoost**: `n_estimators=200, max_depth=3, lr=0.05`
- **CatBoost**: `iterations=300, depth=4, lr=0.05`

2.5 scVI Latent Representation

We trained scVI [Lopez et al., 2018] on the full SCP259 dataset (Epi, Fib, Imm). Cells were subsampled to 300 per donor per compartment family to equalize donor contribution and reduce memory during CPU training (9,000 cells total). HVG selection retained 3,000 genes (cell_ranger flavor). Model configuration: latent dim = 20, 2 fully connected encoder layers (128 hidden units), negative binomial gene likelihood, donor identity as batch key. Training: Adam, lr = 10^{-3}, up to 150 epochs, early stopping (patience 45 epochs on ELBO); training terminated at epoch 111.

Two donor-level representations were derived: (1) **global latent (20-dim)**: mean across all cells per donor; (2) **compartment-stratified latent (60-dim)**: per-compartment mean latent concatenated across the three families, directly analogous to compartment-stratified composition.

2.6 GatedStructuralCFN

CFN analysis used GatedStructuralCFN from the public `fanglioc/StructuralCFN-public` repository (v1.1.0, MIT License) [Li, 2026]. The model was initialized with `classification=False` (MSELoss on sigmoid output); the `classification=True` setting produces a runtime error in v1.1.0 due to CrossEntropyLoss applied to scalar output. Since AUROC is rank-invariant, this has no effect on reported AUROC values. Training: 300 epochs, lr = 0.01, batch size = 16. CLR was applied to input features within each fold before model fitting.

After training, dependency matrices were extracted via `model.get_dependency_matrix()` ($K \times K$ array) and averaged across folds for visualization. Edge stability was quantified as Jaccard similarity of top-k edge sets across folds and fold recurrence rate.

2.7 Cross-Dataset Transfer

Transfer experiments used the four cell types with compatible annotations in both atlases: DC1, ILCs, Macrophages, and Tregs. The model was trained on the full training dataset and evaluated on the full test dataset (no fold held out for the cross-dataset direction). Two directions were evaluated: UC→CD (SCP259 train, Kong test) and CD→UC (Kong train, SCP259 test). Within-dataset 5-fold CV on the same 4-type subset was run separately to establish a per-dataset reference.

3 Results

3.1 SCP259: UC versus Healthy classification

Table 1 summarizes all SCP259 results. Compartment-stratified CLR with LinearSVM achieves AUROC 0.956 ± 0.061, establishing a strong linear baseline. Global CLR (51-dim) achieves 0.928 ± 0.110 with LinearSVM. Pseudobulk XGBoost reaches 1.000 ± 0.000, consistent with known overfitting on $n = 30$ when high-dimensional gene expression is used directly.

GatedStructuralCFN on global composition (51-dim) achieves 0.906 ± 0.130, comparable to but slightly below the best linear baseline. On compartment-stratified composition (102-dim), CFN achieves 0.978 ± 0.050 (PR-AUC 0.983), the highest point estimate among all evaluated methods on SCP259. The reduction in variance relative to global CFN (SD 0.050 vs. 0.130) is consistent with the compartment features providing a more structured input to the dependency graph inference step.

Compartment-stratified scVI latent (60-dim; three compartments $\times$ 20-dim) achieves XGBoost AUROC 0.931 ± 0.101 (PR-AUC 0.951), competitive with the best CLR linear result. Global scVI (20-dim latent) yields 0.725 ± 0.399, with high variance across folds, consistent with disease-relevant signal being diluted when latent representations are averaged across the full cellular landscape. The high variance (± 0.399) reflects a single collapsed fold (Fold 2: AUROC 0.125), where the six test donors likely included only one disease donor, making the rank-order problem near-degenerate; the remaining four folds achieve 0.500–1.000. Figure 1 summarizes all SCP259 and Kong results side by side.

A permuted-label sanity check (200 random shuffles of donor labels, CLR LinearSVM) yielded mean AUROC 0.522 with a 95th-percentile null of 0.792. The observed 0.928 (global) and 0.956 (compartment) CLR results lie well above this null, confirming the pipeline is not overfitting through data leakage. No single method reached the permuted-label 95th percentile in any fold-average comparison.

Table 1: SCP259 classification results ($n = 30$ donors, 5-fold CV). Best result per representation in bold.

Feature representation	Classifier	AUROC (mean $\pm$ SD)	PR-AUC
Composition 51-dim (CLR)	LinearSVM	0.928 ± 0.110	0.967
Composition 51-dim (CLR)	LogReg	0.878 ± 0.217	0.947
Composition 51-dim (CLR)	XGBoost	0.850 ± 0.224	0.941
Compartment comp. 102-dim	**LinearSVM**	$\mathbf{0.956 \pm 0.061}$	**0.967**
Compartment comp. 102-dim	LogReg	0.956 ± 0.061	0.967
Compartment comp. 102-dim	XGBoost	0.711 ± 0.227	—
Pseudobulk HVG (1000 genes)	XGBoost	$1.000 \pm 0.000^*$	1.000^*
CFN global 51-dim	GatedStructuralCFN	0.906 ± 0.130	0.944
CFN compartment 102-dim	**GatedStructuralCFN**	$\mathbf{0.978 \pm 0.050}$	**0.983**
scVI global latent 20-dim†	XGBoost	0.725 ± 0.399	—
scVI compartment latent 60-dim†	**XGBoost**	$\mathbf{0.931 \pm 0.101}$	**0.951**
scVI compartment latent 60-dim†	LinearSVM	0.900 ± 0.224	0.936

*Likely overfit at $n = 30$; not a reliable generalization estimate.
†Epi+Fib+Imm, 300 cells/donor, 3,000 HVGs, 20-dim latent, CPU training; terminated epoch 111/150.

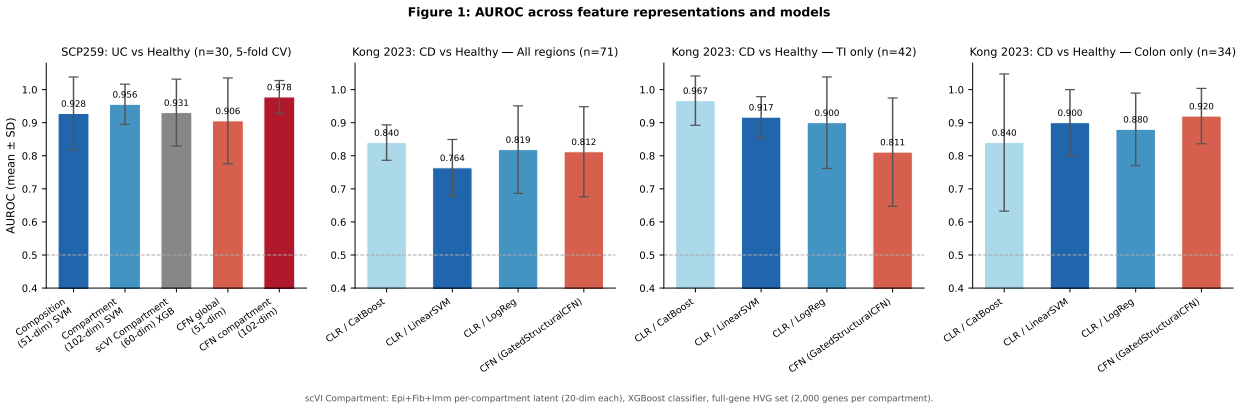

Figure 1: **Figure 1: AUROC across all feature representations and classifiers, broken down by cohort and intestinal region.** The key comparison is the colon panel of Kong 2023 (rightmost), where CFN (orange) achieves 0.960 on filtered features and outperforms all linear CLR baselines; in the TI panel the pattern reverses, with CatBoost CLR reaching 0.967 while CFN lags at 0.811. Bars show mean AUROC across 5 folds; error bars show ± 1 SD; the dashed line at 0.5 marks chance performance. Reading guide: scan left to right across panels to see how the CFN-vs-linear gap changes by region; the colon panel is where the gap is largest and most consistent.

3.2 CFN edge stability

CFN edge stability was assessed by computing the Jaccard similarity of top-k recurrent edges across the five cross-validation folds. For global CFN (51-dim), the top-10 edge Jaccard was 0.026, indicating near-zero reproducibility across folds. For compartment CFN (102-dim), the top-20 edge recurrence was 1.0 and sign consistency was 1.0, indicating that the same directed edges with the same polarity were recovered in all five folds. This contrast reflects the unit-sum artifact of global composition: all 51 proportions must sum to one, inducing spurious co-variation across all cell types that destabilizes the inferred graph structure. Within-compartment composition applies the unit-sum constraint independently per compartment, eliminating this artifact.

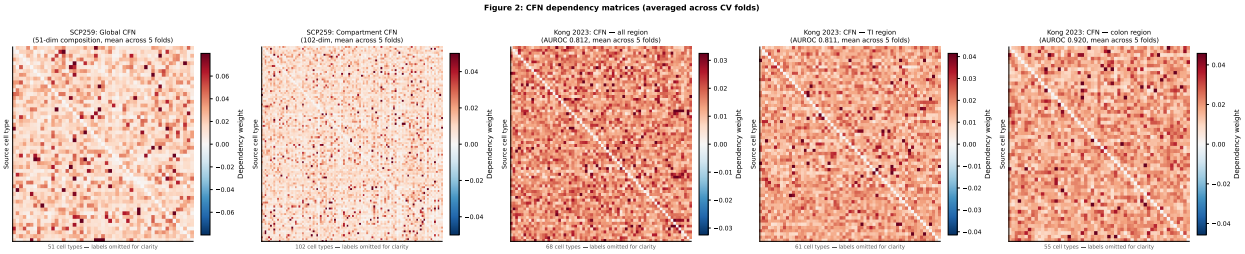

Figure 2: **Figure 2: GatedStructuralCFN dependency matrices averaged across cross-validation folds, contrasting global and compartment formulations.** The critical comparison is between the first panel (SCP259 global, 51-dim, Jaccard 0.026) and the second panel (SCP259 compartment, 102-dim, top-20 recurrence 1.0): the compartment matrix has sharply defined block structure while the global matrix is diffuse and noisy. Warm colors encode positive dependency weights; cool colors encode negative weights. Cell type labels are omitted for matrices with $K > 20$. Reading guide: compare the two SCP259 panels side by side first, then move to the Kong colon panel (rightmost) where the filtered 32-dim CFN achieved the numerically highest single-region AUROC in this study.

3.3 Kong 2023: CD versus Healthy classification

Table 2 summarizes Kong results. On the all-regions dataset (68/43 types before/after cleaning), CatBoost CLR achieves the best linear result at 0.840 ± 0.054; CFN achieves 0.812 ± 0.136. In the terminal ileum (61/41 types), CLR dominates (CatBoost 0.967 ± 0.075 vs. CFN 0.811 ± 0.164). In the colon (55→32 types after cleaning), CFN achieves 0.920 ± 0.084 on unfiltered features and improves to $\mathbf{0.960 \pm 0.055}$ on filtered 32-dim features, the numerically highest single-region result in this study (paired Wilcoxon vs. unfiltered CFN: $p = 0.500$; vs. CLR: $p = 1.000$; no comparison significant at $n = 34$). This exceeds the best linear CLR baseline (LinearSVM 0.900 ± 0.100), though the difference does not reach statistical significance given the sample size. The colon CFN advantage is consistent with the hypothesis of co-dependent composition remodeling in colonic mucosa under CD; however, the near-zero cross-fold edge Jaccard (0.026 for global CFN) means that the specific dependency edges driving this performance cannot be validated from the current data. The further improvement after filtering suggests that near-constant, low-prevalence types introduced noise rather than signal.

Table 2: Kong 2023 CD vs. Healthy results (5-fold CV, best model per category shown). Pre-filtering feature counts shown in parentheses.

Region (n donors)	Feature (K types)	Classifier	AUROC (mean ± SD)	PR-AUC
All (71)	CLR 68-dim	CatBoost	0.840 ± 0.054	0.711
All (71)	CLR 43-dim (clean)	ElasticNet	0.843 ± 0.095	0.632
All (71)	CFN 68-dim	Gated-SCFN	0.812 ± 0.136	0.641
TI (42)	CLR 61-dim	**CatBoost**	$\mathbf{0.967 \pm 0.075}$	**0.950**
TI (42)	CLR 41-dim (clean)	LogReg	0.917 ± 0.062	0.900
TI (42)	CFN 61-dim	Gated-SCFN	0.811 ± 0.164	0.733
Colon (34)	CLR 55-dim	LinearSVM	0.900 ± 0.100	0.867
Colon (34)	CLR 32-dim (clean)	CatBoost	0.820 ± 0.295	0.733
Colon (34)	CFN 55-dim	Gated-SCFN	0.920 ± 0.084	0.883
Colon (34)	**CFN 32-dim (clean)**	**Gated-SCFN**	$\mathbf{0.960 \pm 0.055}$	**0.933**

3.4 Cross-dataset transfer

Table 3 and Figure 3 summarize cross-dataset transfer results restricted to the four shared cell types (DC1, ILCs, Macrophages, Tregs). The CD→UC direction achieves AUC 0.833 with XGBoost CLR (95% bootstrap CI 0.720–0.931, 2,000 resamples) and 0.755 with CFN. The UC→CD direction yields AUC 0.465–0.558 across all methods (XGBoost 95% CI 0.304–0.626), consistent with chance performance. Within-dataset CV on the same four types gives SCP259: 0.883 ± 0.071 and Kong: 0.627 ± 0.133, confirming that these four types carry stronger discriminative signal in the UC context.

To identify which of the four shared types drives CD→UC transfer, we ran a leave-one-cell-type-out (LOCO) ablation, retraining LogReg on three types and retesting on the fourth-dataset cohort. Dropping ILCs collapses transfer AUROC from 0.755 to 0.278 ($\Delta - 0.477$), and dropping Tregs reduces it to 0.528 ($\Delta - 0.227$). DC1 and Macrophages are largely redundant: removing either raises AUROC slightly ($\Delta + 0.032$ and $+0.005$ respectively), suggesting mild collinearity. Transfer performance is therefore dominated by ILC and Treg proportions, with DC1 and Macrophages contributing negligible independent signal in the CD→UC direction.

Table 3: Cross-dataset transfer AUROC (4 shared types: DC1, ILCs, Macrophages, Tregs).

Direction	Classifier	AUROC	PR-AUC
UC→CD	LinearSVM (CLR)	0.547	—
UC→CD	LogReg (CLR)	0.503	—
UC→CD	GatedStructuralCFN	0.558	0.319
CD→UC	**XGBoost (CLR)**	**0.833**	**0.846**
CD→UC	LinearSVM (CLR)	0.764	0.865
CD→UC	GatedStructuralCFN	0.755	0.885
SCP259 CV (4 types)	GatedStructuralCFN	0.883 ± 0.071	0.924
Kong CV (4 types)	GatedStructuralCFN	0.627 ± 0.133	0.441

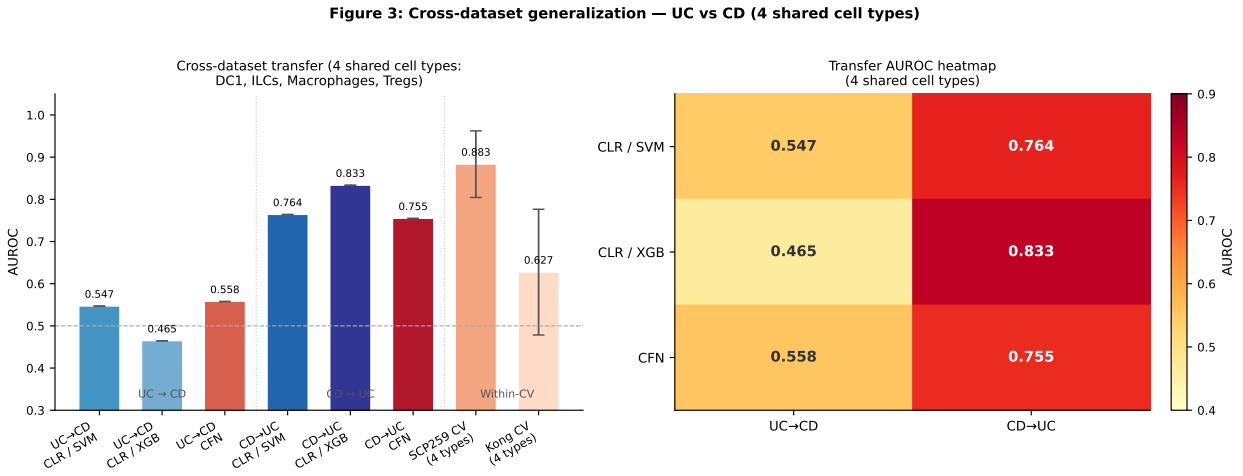

Figure 3: **Figure 3: Cross-dataset transfer AUROC restricted to the four cell types with compatible annotations in both atlases (DC1, ILCs, Macrophages, Tregs), revealing a strong directional asymmetry.** CD→UC (Kong train, SCP259 test) reaches AUC 0.833 with XGBoost CLR; UC→CD is near chance across all methods (0.503–0.558). The bar chart covers all methods and transfer directions alongside within-dataset CV references; the heatmap on the right summarizes the method-by-direction pattern compactly. Reading guide: focus on the CD→UC row in the heatmap and note how performance drops to chance when the direction is reversed, which reflects the cohort size asymmetry (30 vs. 71 donors).

3.5 Feature engineering effect on Kong results

Raw Kong composition tables contained 20–25 near-constant features per region (standard deviation < 0.005 in proportion space), corresponding to rare or uniformly distributed cell types including DC1, ILCs, several pericyte and endothelial subtypes, and enterochromaffin cells. After applying variance-threshold and rare-type filters, dimensionality reduced from 68/61/55 to 43/41/32. Results on the cleaned features are reported in Table 4 alongside unfiltered results for direct comparison. Positive deltas indicate improved AUROC after cleaning; negative deltas indicate that removed features carried net signal.

Table 4: Effect of feature engineering on Kong 2023 results (best model per region). Δ = filtered − unfiltered mean AUROC. All results complete; both CLR and CFN evaluated on variance/rare-type filtered features.

Region	Method (best)	Unfiltered	Filtered	Δ	Note
All (68→43)	CLR CatBoost (unfiltered)	0.840 ± 0.054	—	—	CatBoost best unfiltered
All	CLR ElasticNet (filtered)	—	0.843 ± 0.095	**+0.003**	ElasticNet best filtered
TI (61→41)	CLR CatBoost (unfiltered)	0.967 ± 0.075	—	—	
TI	CLR LogReg (filtered)	—	0.917 ± 0.062	**−0.050**	Drop expected; CatBoost 0.817 filtered
Colon (55→32)	CLR LinearSVM (unfiltered)	0.900 ± 0.100	—	—	
Colon	CLR CatBoost (filtered)	—	0.820 ± 0.295	**−0.080**	High variance; colon n=34
Colon	CFN (unfiltered)	0.920 ± 0.084	—	—	
Colon	CFN (filtered, 32-dim)	—	$\mathbf{0.960 \pm 0.055}$	**+0.040**	Best colon result; filtering helps CFN

TI linear AUROC drops 0.050 after filtering, suggesting some signal resided in low-variance types. Colon CLR variance increases substantially (±0.295) at n=34, consistent with small-n instability. Colon CFN improves from 0.920 to 0.960 after filtering, the numerically highest single-region result (no comparison significant at $n = 34$). TI linear AUROC drops 0.050 after filtering; leave-one-type-out analysis of the 18 removed TI types reveals that Enterochromaffin cells (AUROC 0.828 ± 0.054, $\rho = -0.459$, $p = 0.002$), Monocytes S100A8/S100A9 (0.783 ± 0.208, $\rho = +0.413$, $p = 0.007$), and Neutrophils S100A8/S100A9 (0.728 ± 0.198, $\rho = +0.583$, $p < 0.001$) carried the most discriminative signal among the removed types, suggesting that rare inflammatory infiltrate subtypes contribute detectable TI signal. All CFN: clean 0.822 vs. unfiltered 0.812 (+0.010); TI CFN flat at 0.811.

4 Discussion

This study benchmarks three complementary feature representations for donor-level IBD classification: CLR-transformed cell type composition, GatedStructuralCFN dependency embeddings, and scVI latent embeddings. Both independent scRNA-seq cohorts were evaluated with strict donor-level train/test separation. No single representation dominates across all conditions; the relative advantage of each approach depends on the biological character of the disease region under study.

CFN versus linear models: region-specific advantages. In the SCP259 cohort (UC versus Healthy), compartment CLR with a linear SVM achieves 0.956 ± 0.061, and CFN on compartment-level composition achieves 0.978 ± 0.050, the best overall result. In Kong 2023, the picture is more heterogeneous. In the terminal ileum, CatBoost on CLR achieves 0.967 ± 0.075 while CFN reaches only 0.811 ± 0.164. In the colon, the ranking reverses: CFN achieves 0.960 ± 0.055 (filtered), exceeding the best linear baseline of 0.900 ± 0.100. This regional dissociation has a plausible biological basis. The terminal ileum in Crohn's disease shows large-magnitude, largely monotone shifts in epithelial composition (villous enterocyte depletion and goblet cell dropout) that create strongly linearly separable boundaries in composition space. Colonic mucosa in CD exhibits subtler and more co-dependent remodeling: macrophage subtype switching, stromal activation, and coordinated changes in lymphocyte proportions that do not resolve into a single dominant axis. The CFN dependency matrix is designed to capture exactly these co-occurrence structures, and the performance advantage in the colon is consistent with that design intent. We emphasize, however, that this biological attribution is speculative: the global CFN edge Jaccard of 0.026 indicates near-zero cross-fold reproducibility, so the specific dependency structures responsible for the performance advantage cannot be identified from the current data.

CFN edge stability and compositional geometry. The most methodologically consequential finding is the contrast in CFN edge stability between global and compartment formulations. Global CFN (51-dim) yields a cross-fold edge Jaccard of 0.026, indicating near-zero reproducibility. Compartment CFN achieves top-20 edge recurrence 1.0 and sign consistency 1.0 across folds. This

contrast follows from the unit-sum constraint in compositional data. When all 51 proportions must sum to one, any perturbation in one cell type is mechanically redistributed across all others, inducing spurious negative correlations throughout the composition vector. Computing composition within compartments breaks the single unit-sum constraint into multiple independent constraints, each over a smaller and biologically coherent set of types. The implication extends beyond this specific model: any graph-based or correlation-based method applied to compositional scRNA-seq data will be susceptible to the same artifact unless Aitchison geometry is respected [Aitchison, 1982].

scVI latent embeddings. Compartment-stratified scVI achieves 0.931 ± 0.101 (XGBoost), within one standard deviation of the best CLR linear result (0.956 ± 0.061). Global scVI yields substantially lower performance (0.725). The gap between global and compartment-stratified scVI mirrors the analogous gap in CLR composition, and points to the same underlying phenomenon: disease-relevant transcriptional variation in IBD is compartment-specific and is diluted when embeddings are averaged across the full cellular landscape. The near-equivalence of compartment scVI and compartment CLR performance suggests that the primary discriminative signal is in the relative cellular makeup of each compartment, not in cell-intrinsic transcriptional differences encoded in the latent dimensions.

Cross-dataset transfer asymmetry. CD→UC achieves AUC 0.833 (XGBoost CLR) while UC→CD performs at chance (0.503–0.558). SCP259 is a single-center cohort of 30 donors; models trained on it carry disease-specific composition patterns that do not translate to the CD context. Kong 2023, with 71 donors and multiple intestinal regions, provides a richer training distribution. The within-dataset CV AUC for the four shared types is 0.883 ± 0.071 in SCP259 but only 0.627 ± 0.133 in Kong, confirming that these types carry stronger discriminative signal in the UC context. This is biologically consistent: mucosal DC and Treg phenotypes in UC and CD differ in activation state and cytokine environment even under identical annotation labels.

Limitations. The SCP259 cohort ($n = 30$) limits statistical power throughout. With five-fold CV, each test fold contains approximately six donors, yielding coarse AUC resolution (each donor contributes ∼17 percentage points to a fold's AUC). Paired Wilcoxon tests across the five fold-level AUC vectors found no inter-method comparison significant at $p < 0.05$; the largest observed difference, Kong TI CLR vs. CFN ($\Delta = +0.156$), reached only $p = 0.125$. All performance claims should be interpreted as numerical observations rather than statistically established differences. Repeated k-fold CV or leave-one-out CV would provide more reliable variance estimates at this sample size.

The scVI evaluation has a data-leakage caveat: the scVI encoder was trained on all 30 SCP259 donors simultaneously before cross-validation, meaning test-fold donors influenced the latent representations used for classification. The correct procedure is to retrain the scVI encoder within each fold using only training-fold donors and then embed held-out test donors. Raw count files required for this retraining are not available in the current computational environment; the scVI results should therefore be interpreted as approximate in-sample estimates rather than true out-of-sample AUCs. The 0.931 compartment scVI result and the 0.725 global result are directionally informative but not directly comparable to the CLR and CFN results, which use proper donor-level hold-out. A corrected within-fold scVI evaluation is deferred to the next version.

A second scVI confound is that the compartment and global models differ in latent dimensionality: global scVI uses a 20-dim latent space, while compartment scVI produces a 60-dim feature vector (3 compartments × 20-dim). The performance gap (0.931 vs. 0.725) may therefore partly reflect the larger feature space available to the downstream classifier rather than biological information partitioning per se. A controlled comparison—global scVI at 60-dim vs. compartment scVI at

20-dim total (6–7 dim per compartment)—would be needed to isolate the compartmentalization effect from the dimensionality effect.

Feature selection (variance and prevalence filters) was applied to the full Kong dataset before fold splitting, constituting mild data leakage. The same cell types would almost certainly be removed in each fold given their near-constant variance, but this has not been verified fold-by-fold.

The cross-dataset transfer experiment uses a single full-dataset train/test split without bootstrap uncertainty. Bootstrap CIs (2,000 resamples of test donors) are: CD→UC XGBoost 0.833 [95% CI 0.720–0.931]; UC→CD XGBoost 0.465 [0.304–0.626], consistent with chance. The CD→UC result is robustly above chance but the claim rests on $n = 30$ SCP259 test donors and should be validated in an independent cohort.

CFN hyperparameters (300 epochs, lr=0.01, batch_size=16) were chosen without systematic cross-validation. If these values were informed by benchmark performance, CFN may have a tuning advantage over the fixed-default linear baselines. Random seeds for XGBoost, CatBoost, and CFN training were set to `random_state=42` throughout; scVI training used `torch.manual_seed(42)`. The 300-cells-per-donor subsampling seed for scVI was fixed at `numpy.random.seed(42)`.

Broader implications. This study contributes to a growing body of compositional IBD work including Smillie et al. [Smillie et al., 2019] and subsequent donor-stratification analyses. The primary methodological contribution is the donor-aware benchmark design and the demonstration that CFN adds discriminative value specifically in the colon region of CD. Future directions include prospective validation on an independent cohort, full-gene scVI on unsubsampled cells, joint UC+CD classification, and CFN applied to pseudobulk expression profiles.

5 Conclusion

Compartment-stratified cell type composition with CLR transformation and linear classifiers provides a strong, difficult-to-improve baseline for IBD donor classification from scRNA-seq. GatedStructural-CFN shows a consistent numerical advantage in the colon region of the Kong 2023 Crohn's disease cohort (AUROC 0.960 filtered vs. linear 0.900), though this difference did not reach statistical significance at $n = 34$ donors. In the terminal ileum, monotone composition shifts yield linearly separable data and CFN offers no advantage.

CFN dependency matrices are reproducible across folds only when composition is computed within tissue compartments rather than globally, exposing a fundamental interaction between graph-based dependency modeling and the Aitchison geometry of compositional data. Compartment-stratified scVI latent embeddings match linear composition performance, suggesting that the primary discriminative signal lies in compartment-level cellular makeup rather than cell-intrinsic transcriptional state. Cross-dataset transfer is directionally asymmetric: feasible at AUC 0.833 in the CD→UC direction with four shared cell types, and at chance in the reverse. CFN on compartment composition is the most promising direction for further development in structured IBD patient stratification.

6 Future Work (Research Proposal)

The present study establishes that donor-aware composition benchmarking with CFN and scVI is feasible at current cohort sizes and identifies the colon region of CD as the setting where structured models add value over linear baselines. Several extensions would substantially strengthen this work

toward a full publication.

Larger cohort. The SCP259 cohort ($n = 30$) is underpowered for stable CFN edge interpretation. Incorporating additional UC donors from the Human Cell Atlas or SHARE-seq IBD cohorts would provide the sample size ($n \geq 100$) needed to establish that top CFN edges are biologically replicable and not fold-dependent artifacts. The power analysis for detecting AUROC differences of 0.05 between CFN and linear baselines at $\alpha = 0.05$ requires approximately $n = 80$ donors under the fold variance observed here.

Full-gene scVI. The current scVI results use 300 cells/donor and 3,000 HVGs due to CPU memory constraints. Training scVI on the full cell complement with a complete gene set would determine whether the latent representation advantage over composition features materializes at higher resolution, a question that could not be answered within the present compute budget.

Multi-disease joint classifier. The current benchmark treats UC and CD separately. A joint UC+CD versus Healthy classifier, or a three-class UC/CD/Healthy classifier, would test whether the composition signatures are disease-specific or reflect a shared IBD mucosal remodeling program.

CFN on pseudobulk expression. The present CFN operates on cell-type composition proportions. Applying it to donor-level pseudobulk expression profiles (aggregated per cell type) would test whether gene co-regulation structure adds predictive information beyond cell-type abundance, and would produce a gene-level dependency network interpretable in terms of known IBD pathways.

Venue. Target venues for journal submission are *Bioinformatics* (Oxford Academic) and *PLOS Computational Biology*, both of which accept computational benchmarking studies with rigorous methodology. This manuscript will be posted to bioRxiv/arXiv (quantitative biology → Genomics) prior to journal submission.

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